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Microorganisms

- Minute living things
- Can not be seen with the naked eye

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Examples

- Protozoans
- Bacteria
- Viruses
- Archae
- Algae
- yeast

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Evolution of Microorganisms

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- Microorganisms came into being on earth over a period of about 1.2 to 1.5 billion years
- Fossil microbes have been found in rocks 3.3 to 3.5 billion years old. Since then, microorganisms have had the principal task of recycling organic matter in the environment.

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Microorganisms

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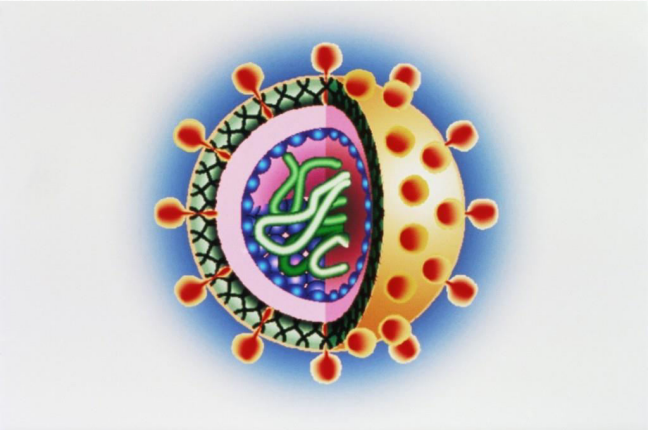
viruses



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Fungi

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protozoa

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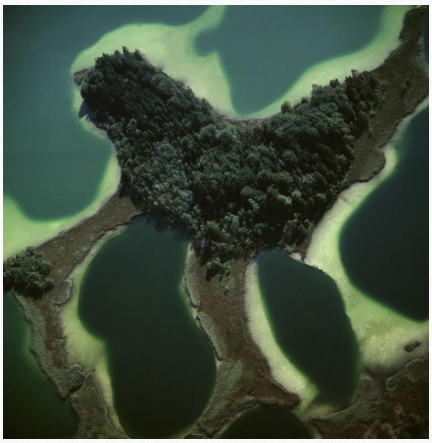
Algae



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Archaea

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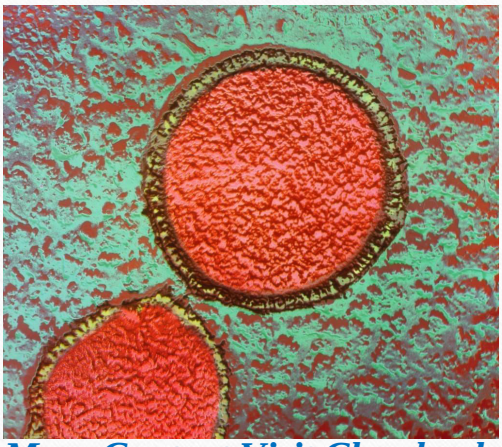

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Usefulness of microbes

- Most of the microbes are either
- Necessary for life
- Good for us
- Can be used for our benefit in the industry

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microbe plant interaction

- Many microbes are fond in nature and help plants to grow
- Rhizobacteria found in the soil fixate nitrogen which is required to grow many crops

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recycling of masses

- microorganisms are responsible for recycling the huge masses of organic matter synthesized by plants as life on earth evolved
- After evolutionmicroorganisms have had the principal task of recycling organic matter in the environment

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Oxygen production

- Cyanobacteria or blue green algae produce oxygen in the ocean
- And this oxygen production is very necessary for aquatic life survival

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decomposition

- Decomposition – Defined as the breakdown of raw organic materials to a finished compost
- The fungi invade the organic matter in soils first and are then followed by bacteria.
- Without this recycling of inorganic nutrients, primary productivity on the globe would stop.

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Industrial use

- **In food industry**
- Cheese and yogurt
- Bread and dough products
- Alcohol production
- **Medicine industry**
- For the production of medicine
- Vaccine production
- Insulin production

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What is biotechnology

- A technique that uses living organisms, or parts of such organisms, to make or modify products, to improve plants or animals, or to develop microorganisms for specific use.

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What is biotechnology

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- Biotechnology is not a separate science but rather a mix of disciplines (genetics, molecular biology, biochemistry, embryology, and cell biology)

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key components of modern biotechnology

- **Genomics:**
- The molecular characterization of all genes in a species.
- **Bioinformatics:**
- The assembly of data from genomic analysis into accessible forms, involving the application of information technology to analyze and manage large data sets resulting from gene sequencing or related techniques.

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key components of modern biotechnology

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- **Transformation:**
- The introduction of one or more genes conferring potentially useful traits into plants, livestock, fish and tree species.
- Genetically improved organism.
- Genetically modified organism (GMO).
- Living modified organism (LMO)

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key components of modern biotechnology

- **Molecular breeding:**
- Identification and evaluation of useful traits in breeding programs by the use of marker-assisted selection (MAS);
- **Diagnostics:**
- The use of molecular characterization to provide more accurate and quicker identification of pathogens

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key components of modern biotechnology

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- **Vaccine technology:**
- The use of modem immunology to develop recombinant deoxyribonucleic acid (rDNA) vaccines for improved control of livestock and fish diseases.

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Introduction to microbial biotechnology

- Application of scientific and engineering principles to the processing of materials by microorganisms to create useful products or processes
- Microorganisms utilized may be natural isolates, laboratory selected mutants or microbes that have been genetically modified using recombinant DNA methods

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Introduction to microbial biotechnology

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- Deals with the prevention of deterioration of processed or manufactured good, environmental protection and waste disposal system
- Production of antibiotics, organic acids and enzymes by fermentation of microbes.

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- Microbial biotechnology, enabled by genome studies, will lead to breakthroughs such as improved vaccines and better disease-diagnostic tools, improved microbial agents for biological control of plant and animal pests, modifications of plant and animal pathogens for reduced virulence, development of new industrial catalysts and fermentation organisms, and development of new microbial agents for bioremediation of soil and water contaminated by agricultural runoff

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- Microbial genomics and microbial biotechnology research is critical for advances in food safety, food security, biotechnology, value-added products, human nutrition and functional foods, plant and animal protection, and furthering fundamental research in the agricultural sciences.

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Scope of microbial bio technology

- **Microorganisms in food**
- Microorganisms are used in the production of fermented food and beverages. E.g. Yeast,*Penicillium, Lactobacillus.*

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Microorganisms in food

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Scope of microbial bio technology

- **Microorganisms in Medicine**
- They also have immense potential in the field of medicine. They are used industrially for the production of antibiotics, vaccines. Insuline, growth hormones and diagnostic kits. E.g. *E. coli,* Polio virus.

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Microorganisms in Medicine

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Scope of microbial bio technology

- **Microorganisms in Industry**
- Industrially important acids, enzymes, pigments are produced with the help of microorganisms. E.g. *Aspergillus niger, Bacillus subtilis.*

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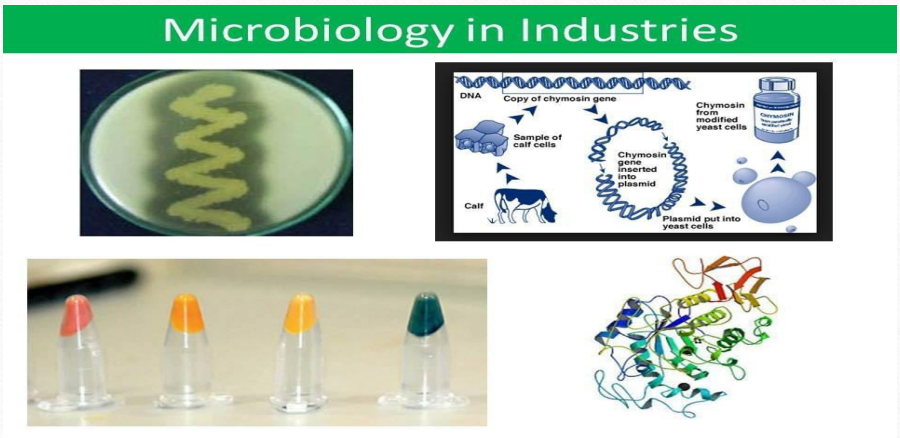

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Scope of microbial bio technology

- **Microorganisms in Agriculture**
- Microorganisms have been ruling in agriculture from the past century. They are used as Biofertilizers and Biopesticides. E.g. *Rhizobium, Bacillus, Azotobacter.*

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Microorganisms in Agriculture

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Scope of microbial bio technology

- **Microorganisms in Industry**
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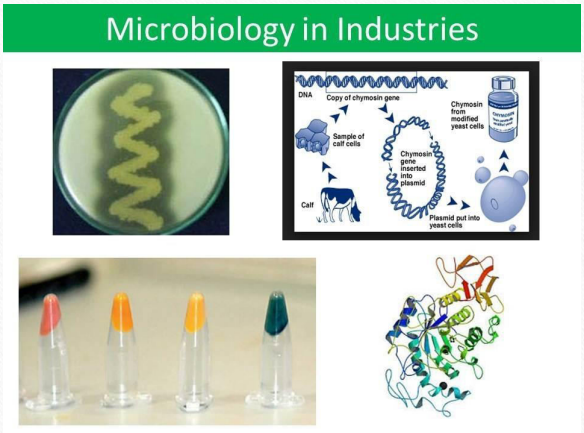

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Scope of microbial bio technology

- **Microorganisms in Environment**
- Ability of microorganisms to degrade toxic materials like oil, petroleum, plastic, etc has opened a new field of research. E.g. *Pseudomonas, Alcanivorax.*

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Microorganisms in Environment

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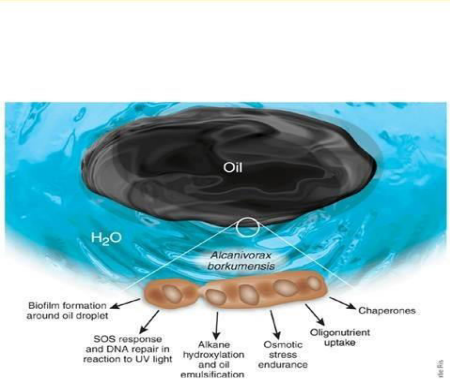
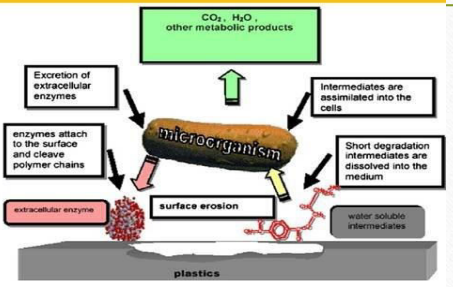
Degradation of toxic waste by Microbes





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Issues with Microbial biotechnology

- Bio processing of microbes
- Application of natural or genetically manipulated whole cells/ tissues/ organs, or parts thereof, for the production of industrially or medically important products
- Encompasses research, development, manufacturing, and commercialization of products prepared by using biological systems

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Issues of microorganisms in bio processing

1st Step:

- Isolation of Pure or Mixed Cultures from nature

2nd Step:

Assesment to determine the capacity for the production

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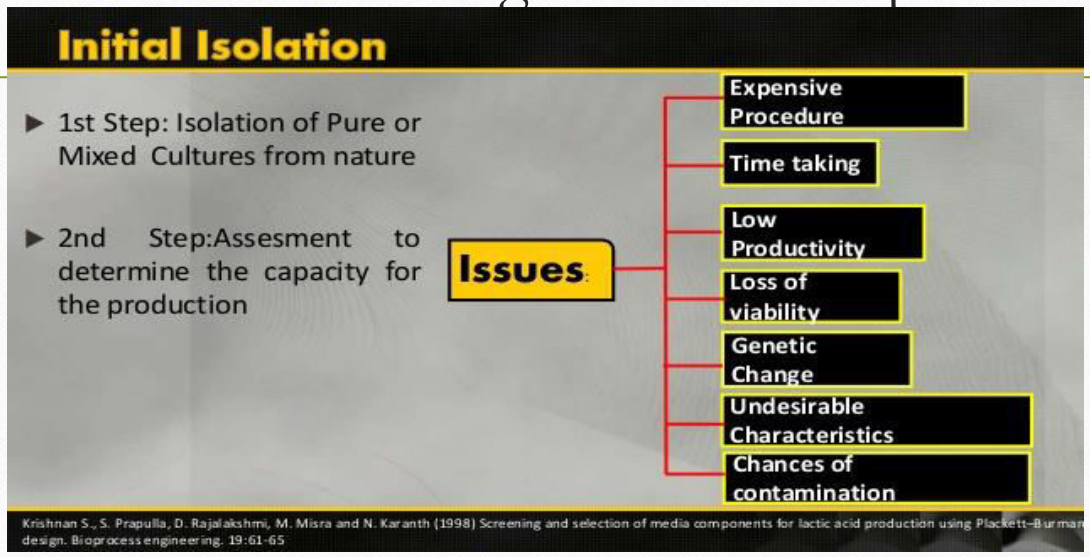
Issues of microorganisms in bio processing



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Issues with Microbial biotechnology II

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- **Strain improvement**
- Selection of natural variants
- Selection of induced mutants
- Use of recombinant technology

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Issues with strain improvement

- May require greater aeration
- Products may pose new extraction challenges
- May even require an entirely new fermentation medium
- Expensive & Laborious
- Need more intensive foam control Issues:

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Issues with strain production

- Physiological Factors
- Physical Factors
- Nutritional Factors

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Physiological Factors

- Morphology
- Metabolic by-products
- Shear Sensitivity
- Genetic Stability
- Issues in production strain:

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Issues with Microbial biotechnology

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- **Physical factors**
- Temperature
- Humidity
- Ph
- viscosity

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Nutritional factors

- Macronutrients
- Micronutrients

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Nutritional factors

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- Nutritional requirements can be determined from chemical composition of microbial cells.

- Improper nutrients will lead to the formation of abnormal molecules and hinders the growth of microorganisms

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Genetically modified organisms GMOs

- **Genetically modified organism (GMO),** organism whose genome has been engineered in the laboratory in order to favour the expression of desired physiological traits or the production of desired biological products.

- A GMO (genetically modified organism) is the result of a laboratory process where genes from the DNA of one species are extracted and artificially forced into the genes of an unrelated plant or animal.

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Genetically modified organisms GMOs

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- GMOs, are commonly used in foods and medicines.

- This has led to concern about the dangers they might cause to the environment and to human

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Food GMOs

Food GMO's

- Crops are modified so when farmer kill weeds with herbicides the crops can with stand the exposure to the herbicide – killing the weeds and not the crop
- Although it is not as common, some types of GMOs are modified to increase their nutrient content. Corn and soybeans are two examples of crops that have higher-nutrient GMO versions available.

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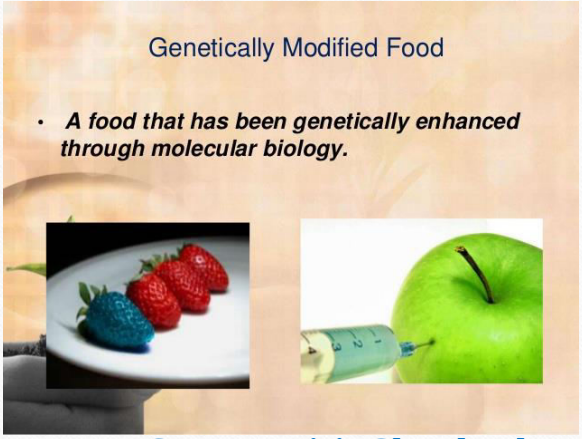
Food GMOs



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Genetically modified organisms GMOs

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Genetically modified organisms GMOs

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Medicine GMO's

- Genetically modified medicines can be produced cheaper and easier.
- Some GMO's are : insulin, thyroid hormones and the hepatitis B vaccine (insulin being the oldest)

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Production of GMOs

- **Production is done by**
- Genetic modification
- **involves the**
- mutation
- insertion
- or deletion of genes.

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Production of GMOs

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- Inserted genes usually come from a different species in a form of horizontal gene transfer
- In nature this can occur when exogenous DNA penetrates the cell membrane for any reason

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Production of GMOs

- **This can be accomplished artificially by**
- attaching the genes to a virus.
- physically inserting the extra DNA into the nucleus of the intended host with a very small syringe.
- using electroporation (that is, introducing DNA from one organism into the cell of another by use of an electric pulse).
- firing small particles from a gene gun

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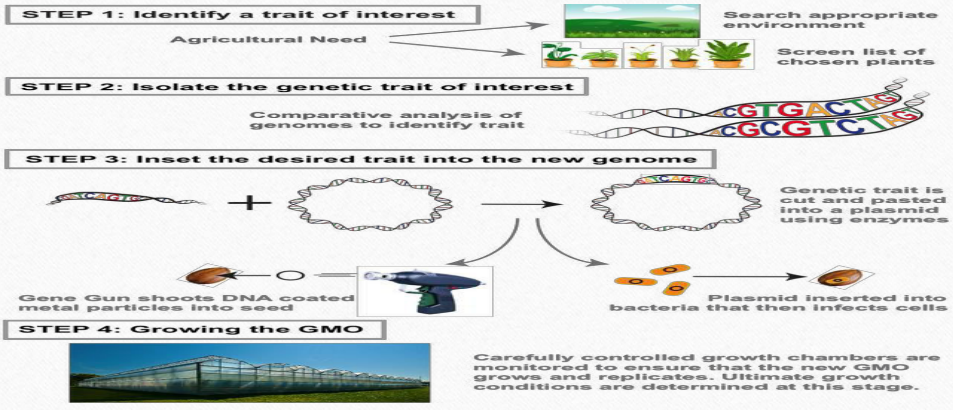
Production of GMOs



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Examples of GMOs

- Plants
- In agriculture, genetically engineered crops are created to possess several desirable traits,
- such as resistance to pests,
- herbicides,
- or harsh environmental conditions,
- improved product shelf life,
- increased nutritional value,
- or production of valuable goods such as drugs (pharming). •

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GM microbes

- Bacteria were the first organisms to be modified in the laboratory, due to their simple genetics. •
- These organisms are now used for several purposes,
- particularly important in producing large amounts of pure human proteins for use in medicine.
- • Genetically modified bacteria are used to produce the protein insulin to treat diabetes.
- • Similar bacteria have been used to produce clotting factors to treat haemophilia,
- and human growth hormone to treat various forms of dwarfism

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Examples of GM plants

- Plants, including algae, jatropha, maize, and other plants have been genetically modified for use in producing fuel, known as biofuel.
- The most common types of GMO's plants are
- Foods – canola ,
- soya bean ,
- golden rice
- Sugar beets
- alfalfa

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Examples of GMOs

- Animals:
- Genetically modified mammals are an important category of genetically modified organisms
- The process of genetically engineering animals is a slow, tedious, and expensive process. However, new technologies are making genetic modifications easier and more precise.

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Most common Examples of GM animals

- Glow fish
- Dolly sheep
- See through frog
- Sudden-Death Mosquitoes

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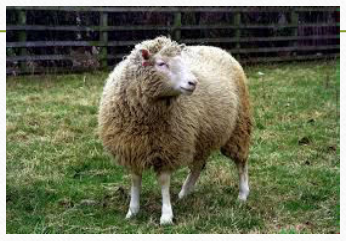
Conti…



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WHAT ARE THE ADVANTAGES OF GMOS

- GMOs is an effective way to provide farmers a larger profit while spending less time a resources.
- Economically beneficial because they are used to repel pests, which prevents the need for pesticides to be used.
- This is also known to decrease food prices due to advanced crops and lower cost. GMOs will certainly help families that cannot afford to buy the food they need for everyday living.

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WHAT ARE THE ADVANTAGES OF GMOS

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- Less starvation in the world due to the fact that it cuts food prices.
- It is more nourishing to the body, which is proven to be effective.
- The precise evaluation and testing of GMOs crops and other products. In fact, according to research, it is safer as compared to traditional crops.

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WHAT ARE THE ADVANTAGES OF GMOS

- Reduce use of pesticide and other Toxic chemicals
- Desired characteristics of food are achieved and in a shorter time
- Improves nutritional value.
- Many people rely on GM food for medicines.
- Gene technology is the best solution to the problem of world hunger.

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Disadvantages of GMOs

- Unacceptable to some groups to copy genes.
- Interfiere in food chain
- Genetically engineered animals may suffer more health problems
- Allergic reactions
- "Genetic Pollution"

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Disadvantages of GMOs

- GMOs could be dangerous to some insects. This is because the new genes of the crops can be deadly to certain insects like butterflies that are not actually dangerous to crops.
- The people who oppose GMOs aren't taste good as compared to naturally produced crops.
- It doesn't need enough pesticide because the crop itself is dangerous to some insects..

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Disadvantages of GMOs

- The tariff, quota and trade issues may become a problem in regions and other countries.
- Critics claim that GMOs can cause particular disease or illnesses.
- As the major producer of GMO, it could start an issue in some population that are not agree in American way

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Disadvantages of GMOs

- Genetically modified crops may also cause a threat to the environment. This is because it is not a natural way to plant and cultivate crops.
- Possible greed or self-indulgence of the manufacturers and companies of GMOs. This is due to the profit that can be acquired.

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Microbes as tool for research

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- Microorganisms are essential for the earth to function
- They play many roles both on land and in water,
- including being the first to colonize and ameliorate effects of naturally occurring and man-made disturbed environments.

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Microbes as tool for research

- Learning more about these microorganisms will be of value for the following reasons:
- microorganisms are important sources of knowledge about the strategies and limits of life,
- microorganisms are of critical importance to the sustainability of life on our planet,
- the untapped diversity of microorganisms is a resource for new genes and organisms of value to biotechnology,

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Microbes as tool for research

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- diversity patterns of microorganisms can be used for monitoring and predicting environmental change,
- microorganisms play a role in conservation and restoration biology of higher organisms, and
- microbial communities are excellent models for understanding biological interactions and evolutionary history.

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Importance of microbes

- **Decayingprocess**
- Manybacteria help decompose (menguraikan) deadorganism and animal wastes into chemical compound such as
- ammonia.
- In sewagetreatment plant (lojirawatankumbahan), bacteria are use to break down the complex animal and plant matter into simple compounds

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IN DIGESTION

- ONE OF THE MAIN COMPONENT OF PLANT IS CELLULOSe
- Cellulose is hard to digest by the animal…
- How they digest the cellulose…?
- Cellulose can be digest by enzyme named cellulase.
- How they can get cellulase?
- Herbivores depend on bacteria that live in their stomach to digest cellulose
- This bacteria secretes (merembeskan) cellulase into the stomach

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MEDICINE

- Some type of microorganism produce important drugs called ANTIBIOTICS.
- Functions of antibiotics….
- Weaken, Destroy other microorganism
- Penicillium notatumis a GREEN MOULD
- Its produces antibiotic called PENICILLIN

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Genetically engineered bacteria

- Some bacteria have been genetically engineered
- Why?
- To produce certain vaccines that are used to prevent infectious diseases
- Insulin, produced to treat diabetes

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AGRICULTURE

- Bacteria play important roles to promote the plant growth.
- Bacteria, Break down dead plants and dead animals by releasing AMMONIA
- AMMONIA, Adds nitrogen into the soil
- The nitrogen fixing bacteria in the soil change the ammonia into nitrates.
- This nitrates will be used by plants

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importance of microbes

- Why plants need nitrates
- they need nitrates to make amino acids
- Pea and bean plants use bacteria Rhizobiumwhich directly convert nitrogen into acid amino

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importance of microbes

- INDUSTRY
- Bacteria are used in the production of
- Milk
- Yogurt
- Cheese
- Vinegar
- Soy sauce
- Chocolate
- Certain vitamin B
- Citric acids

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- Production of bread
- Yeast causes the bread to rise by producing carbon dioxide from the carbohydrates in the dough.
- Carbon dioxide gas bubbles released in the dough causes it to rise.
- So the bread become fluffy

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Microbes and biotechnology

- Microbial biotechnology, enabled by genome studies, will lead to breakthroughs such as
- improved vaccines and better disease-diagnostic tools,
- improved microbial agents for biological control of plant and animal pests, modifications of plant and animal pathogens for reduced virulence,
- development of new industrial catalysts and fermentation organisms, and
- development of new microbial agents for bioremediation of soil and water contaminated by agricultural runoff

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Microbes and biotechnology

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- Microbial biotechnology is an important area that promotes for advances in food safety,
- food security,
- value-added products,
- human nutrition
- and functional foods,
- plant and animal protection,
- and overall fundamental research in the agricultural sciences.

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Microbes and biotechnology

- These organisms live in soil, water, food, animal intestines and other different environments.
- Various microbial habitats reflect an enormous diversity of biochemical and metabolic traits
- that have arisen by genetic variation and natural selection in microbial populations.

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Microbes and biotechnology

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- Men used some of microbial diversity
- in the production of fermented foods such as bread, yogurt, and cheese.
- Some soil microbes release nitrogen that plants need for growth and emit gases
- that maintain the critical composition of the Earth's atmosphere.

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Microbes and biotechnology

- Other microbes challenge the food supply by causing yield-reducing diseases in food-producing plants and animals.
- In our bodies, different microbes help to digest food, ward off invasive organisms, and engage in skirmishes and pitched battles with the human immune system in the give-and-take of the natural disease process.

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Significance of microbes in food production

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- Microorganisms, particularly the bacteria and fungi, have served humans since hundreds of years for the purpose of food
- Bacteria are used to make a wide range of food products. The most important bacteria in food manufacturing are Lactobacillus species, also referred to as lactic bacteria.

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Dairy industry

- It would be impossible to make **cheese** without a starter culture
- The production of cheese utilizes starter cultures being added to milk with the aim of producing semi-hard and hard cheeses
-
- The starter cultures produce lactic acid, resulting in a pH decrease in the milk and promoting curd synthesis

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Dairy industry

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- **Yogurt** is also a fermented milk product, which takes advantage of lactic acid generating bacteria cultures Lactobacillus and their use of lactose in milk as a substrate to produce lactic acid
- In **yoghurt** and other fermented milk products, the culture is responsible for the taste and texture of the final product.

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Conti…

- In recent years, **probiotic cultures** have become popular in dairy products because of their health benefits.
- These cultures are all very carefully selected strains, and there is good evidence that they help improve digestion, safeguard the immune system, and keep the body's intestinal flora in balance.

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Meat industry

- Meat starter cultures are used to make dried, fermented products
- such as salami, Salami is a type of cured sausage consisting of fermented and air-dried meat, typically beef or pork.
- pepperoni, Pepperoni is an American variety of salami, made from cured pork and beef mixed together and seasoned with paprika or other chili pepper.
- dried ham, **Ham** is pork that has been preserved through salting, smoking, or wet curing

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Meat industry

- Lactic bacteria develop the flavour and colour of the products.
- In addition, a wide variety of moulds are used
- to ripen the surface of sausages,
- preserving the natural quality of the product
- and controlling the development of flavour

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Meat industry

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Wine industry

- Yeasts are responsible for the fermentation process which produces alcohol in wine.
- However, lactic bacteria also play an important role, as they convert the unstable malic acid that is naturally present in wine into the stable lactic acid.
- This conversion gives the stability that is characteristic of high-quality wines that improve on storage.

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List of microorganisms used in food and beverage preparation

microorganism	Type of microorganism	Food or beverage
Acetobacter aceti	bacterium	chocolate
Acetobacter aceti	bacterium	vinegar
Acetobacter cerevisiae	bacterium	Beer
Acetobacter fabarum	bacterium	chocolate
Acetobacter fabarum	bacterium	coffee
Acetobacter lovaniensis	bacterium	vegetables
Acetobacter malorum	bacterium	vinegar

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Conti…

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microorganism	Type of microorganism	Food or beverage
Kluyveromyces lactis	fungus	cheese
Kluyveromyces marxianus	fungus	cheese
Rhodotorula glutinis	fungus	chocolate
Rhodotorula rubra	fungus	chocolate
Saccharomyces cerevisiae	fungus	chocolate

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Fermentation

- **Fermentation** is a metabolic process that consumes sugar in the absence of oxygen
- **Fermentation** is the conversion of carbohydrates (plant foods) to alcohols and carbon dioxide, or organic acids, using yeasts, bacteria, or a combination of that, under anaerobic (no oxygen) conditions

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Fermentation

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In microorganisms, fermentation is the primary means of producing energy by the degradation of organic nutrients anaerobically.

Humans have used fermentation to produce drinks and beverages since the neolithic age

For example, fermentation is used for preservation in a process that produces lactic acid as found in such sour foods as pickled cucumbers and yogurt

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Fermentation

- The term fermentation sometimes refers specifically to the chemical conversion of sugar into ethanol producing alcoholic drinks such as wine beer and cider.
- Fermentation can even occur within the stomachs of animals, including humans.

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Types of fermentation

- Two types of fermentation
- Lactic Acid
- Alcoholic

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Lactic acid fermentation

- occurs in animal muscles when the tissue requires energy at a faster rate than oxygen can be supplied
- used to convert lactose into lactic acid in yogurt and cheese production
- also gives the sour taste to fermented vegetables such as pickles

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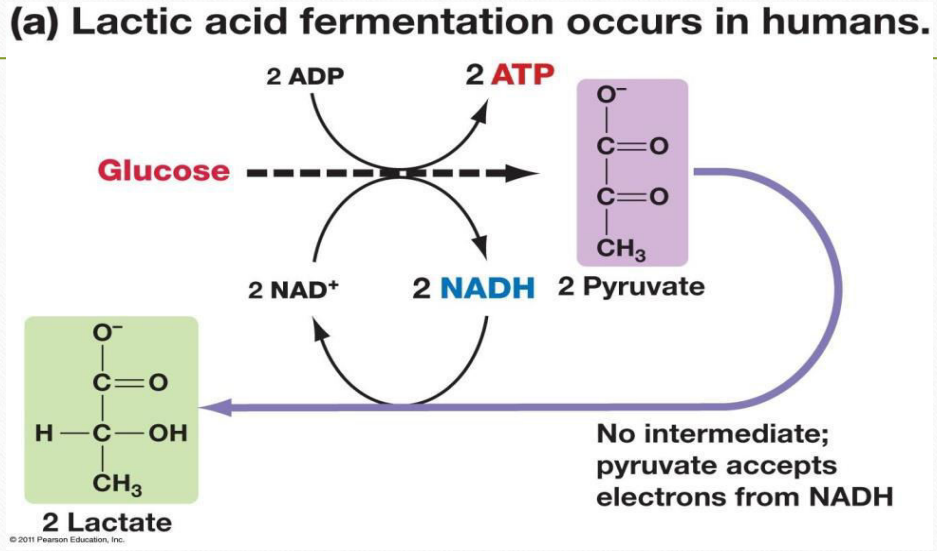
Lactic acid fermentation



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Alcoholic fermentation

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- Done by
- Yeast and certain bacteria
- Pyruvate is broken into alcohol and carbon dioxide
- Used in the production of beer, wine and bread

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Alcoholic fermentation

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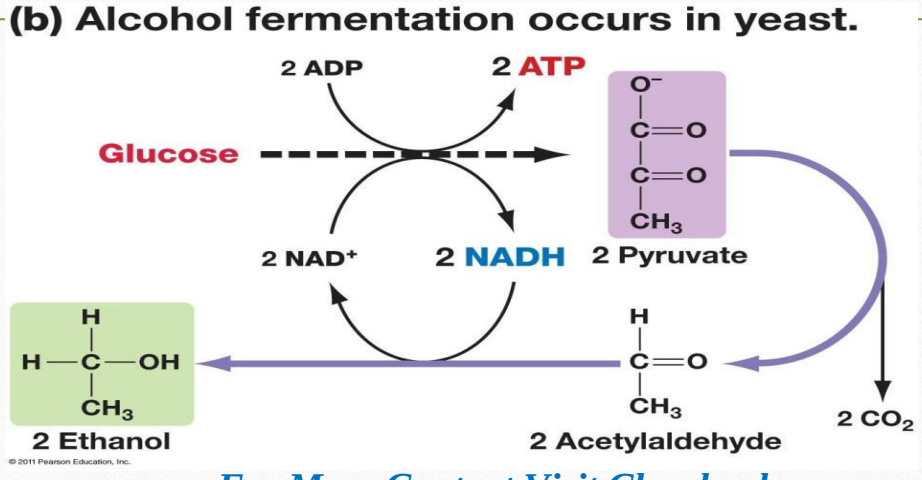
Alcoholic fermentation



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Role of microorganisms in industries

- Microbes are widely used in large-scale industrial processes
- microbes are used to synthesise a number of products valuable to human beings.
- This Industry has provided products that have profoundly changed our lives and life spans.
- These products include beverages, food additives, products for human and animal health, and biofuels.

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Beverages

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- Microbes especially yeast have been used from time immemorial for the production of beverages like **Wine**, **Beer**, **Whisky**, **Brandy**or **Rum**.
- For this purpose, the yeast *Saccharomyces cerevisiae*(commonly called **Brewer's yeast**) is used for fermenting malted cereals and fruit juices to produce ethanol.

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Antibiotics

- **Antibiotics** produced by microbes are regarded was one of the most significant discoveries of the twentieth century and have made major contributions towards the welfare of human society.

- Many antibiotics are produced by microorganisms, predominantly by *Actinomycetes* in the genus Streptomycin (e.g. **Tetracycline, Streptomycin, Actinomycin D**) and by filamentous fungi (e.g. **Penicillin, Cephalosporin**).

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Organic acids

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- Microbes are also used for the commercial and industrial production of certain organic acids. These compounds can be produced directly from glucose (e.g. **gluconic acid**) or formed as end products from pyruvate or ethanol.

- Examples of acids producers are *Aspergillus Niger* (**a fungus**) of Citric acid, *Acetobacter acute* (**a bacterium)** of acetic acid, *Lactobacillus* (**a bacterium)** of lactic acid and many others.

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Enzymes

- Many microbes synthesise and excrete large quantities of enzymes into the surrounding medium. Using this feature of these tiny organisms, many enzymes have been produced commercially.
- These include **Amylase, Cellulase, Protease, Lipase, Pectinase, Streptokinase** and many others.
- Enzymes are extensively used in **food processing and preservation, washing powders, Leather Industry, Paper Industry** and in scientific research

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Vitamins

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- **Vitamins** are some organic compounds which are capable of performing many life-sustaining functions inside our body.
- These compounds cannot be synthesised by humans, and therefore, they have to be supplied in small amounts in the diet.
- Microbes are capable of synthesizing these compounds and hence they can be used for the commercial production of many of the Vitamins e.g. **thiamine (Vitamin $B_{1)}$, riboflavin (Vitamin B_2)**

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Biofuels

- Organic solvents such as **ethanol, acetone, butanol** and **glycerol**are some very important chemicals that are widely used in petrochemical industries.
- These chemicals can be commercially produced by using microbes and low-cost raw materials (e.g. **wood, cellulose, starch**).

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Steroids

- **Steroids** are a very important group of chemicals, which are used as anti-inflammatory drugs, and as hormones such as ***estrogens*** and ***progesterone***, which are used in oral contraceptives.
- Producing steroids from animal sources or chemically systhesising them is difficult, but microorganisms can synthesise steroids from **sterols** or from related compounds.

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Vaccines

- **Vaccines** are a product of **Industrial Microbiology**. Many antiviral vaccines are mass-produced in chicken eggs or cell cultures.

- The production of vaccines against bacterial diseases usually requires the growth of large amounts of the bacteria. **Recombinant DNA technology** is increasingly important in the development and production of subunit vaccines.

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Pharmaceutical Drugs

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- Many pharmaceutical drugs are also produced by microbes e.g. **Cyclosporin A**, that is used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus *Trichoderma polysporum*.

- **Statins** produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol-lowering agents. It acts by competitively inhibiting the enzyme responsible for the synthesis of **cholesterol**.

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introduction

- A vaccine is a biological preparation that improves immunity to a particular disease.
- A vaccine typically contains an agent that resembles a disease-causing microorganism and is often made from weakened or killed forms of the microbe.

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Introduction

• The agent stimulates the body's immune system to recognize the agent as foreign, destroy it, and keep a record of it.

• So that the immune system can more easily recognize and destroy any of these microorganisms that it later encounters.

• The terms vaccine and vaccination are derived from Variolae vaccinae (smallpox of the cow), the term devised by Edward Jenner to denote cowpox.

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History of vaccine

- During the late 1760s whilst serving his apprenticeship as a surgeon Edward Jenner learned of the story, common in rural areas, that dairy workers would never have the often-fatal or disfiguring disease smallpox
- Because they had already had cowpox, which has a very mild effect in humans.

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History of vaccine

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- In 1796, Jenner took pus from the hand of a milkmaid with cowpox, scratched it into the arm of an 8-year-old boy.
- Six weeks later inoculated the boy with smallpox, afterwards observing that he did not catch smallpox.
- Jenner extended his studies and in 1798 reported that his vaccine was safe in children and adults.

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- In 1885 the French microbiologist Louis Pasteur and Emile roux developed the first vaccine against **Rabies**

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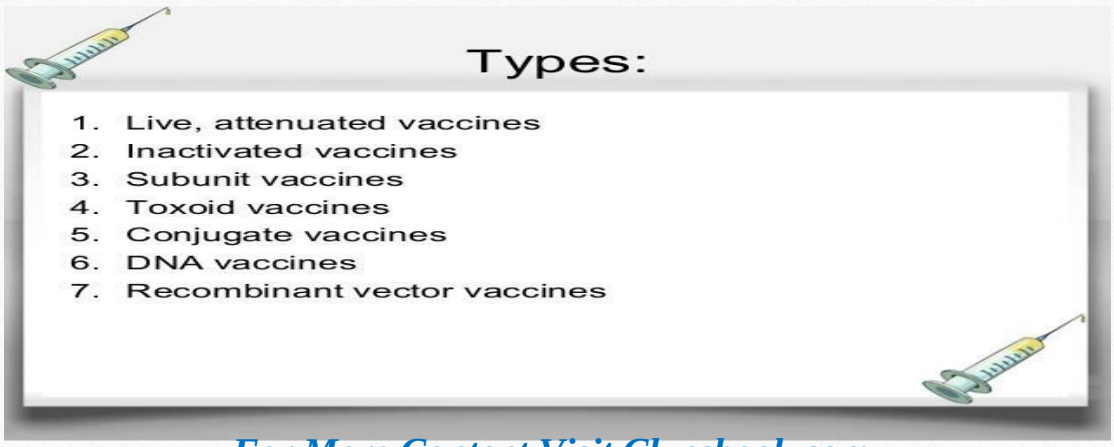
Types of vaccine



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Live, Attenuated Vaccines

• Live, attenuated vaccines contain a version of the living microbe that has been weakened in the lab so it can't cause disease.

• Because a live, attenuated vaccine is the closest thing to a natural infection, these vaccines are good "teachers" of the immune system.

• Example: Vaccines against measles, mumps, and chickenpox

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Inactivated vaccines

- Scientists produce inactivated vaccines by killing the disease-causing microbe with chemicals, heat, or radiation. Such vaccines are more stable and safer than live vaccines.
- Because dead microbes can't mutate back to their disease-causing state.
- Example: Vaccines against influenza, polio, hepatitis A, and rabies.

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Subunits Vaccines:

• Instead of the entire microbe, subunit vaccines include only the antigens that best stimulate the immune system.

• In some cases, these vaccines use epitopes the very specific parts of the antigen that antibodies or T cells recognize and bind to. immunization.

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- Because subunit vaccines contain only the essential antigens and not all the other molecules that make up the microbe.

• Example: Plague immunization.

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Toxoid Vaccines

- For bacteria that secrete toxins, or harmful chemicals, a toxoid vaccine might be the answer.
- These vaccines are used when a bacterial toxin is the main cause of illness.
- Scientists have found that they can inactivate toxins by treating them with formalin.

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- Such "detoxified" toxins, called toxoids, are safe for use in vaccines.
- Example:

Crotalus atrox toxoid is used to vaccinate dogs against rattlesnake bites.

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Conjugate Vaccines

- If a bacterium possesses an outer coating of sugar molecules called polysaccharides, as many harmful bacteria do, researchers may try making a conjugate vaccine for it.

• Polysaccharide coatings disguise a bacterium's antigens so that the immature immune systems of infants and younger children can't recognize or respond to them.

• Example : Haemophilus influenzae type B vaccine.

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DNA Vaccines:

- Still in the experimental stages, these vaccines show great promise, and several types are being tested in humans.
- • DNA vaccines take immunization to a new technological level.
- • These vaccines dispense with both the whole organism and its parts and get right down to the essentials: the microbe's genetic material.
- • Example: Influenza vaccine.

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Recombinant Vector Vaccines

- Recombinant vector vaccines are experimental vaccines similar to DNA vaccines

• But they use an attenuated virus or bacterium to introduce microbial DNA to cells of the body. • "Vector" refers to the virus or bacterium used as the carrier.

• Example : DPT (The vaccine components include diphtheria and tetanus toxoids and killed whole cells of the organism that cause pertussis)

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adjuvant

- An adjuvant is a substance that is formulated as part of a vaccine to enhance its ability to induce protection against infection.

- The word "adjuvant" comes from the Latin *adjuvare* and means "to help."
- Adjuvants help activate the immune system, allowing the antigens—pathogen components that elicit an immune response—in vaccines to induce long-term protective immunity.

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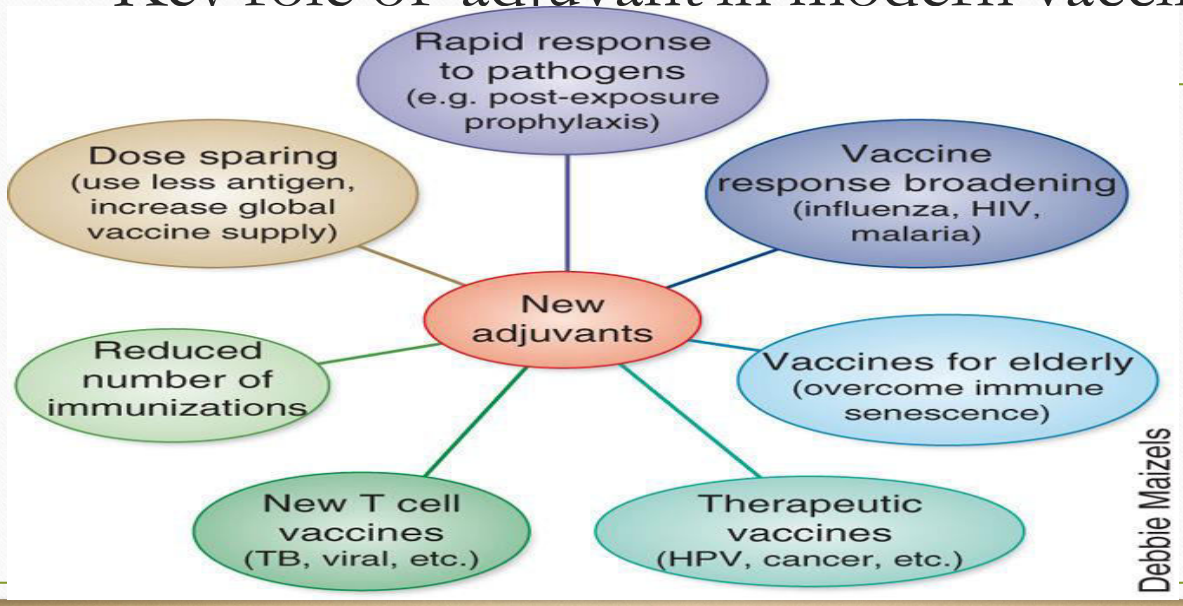
Kev role of adiuvant in modern vaccine



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Challegnes with adjuvant

- Vaccines containing novel adjuvant formulations are increasingly reaching advanced development and licensing stages, providing new tools to fill previously unmet clinical needs.
- However, many adjuvants fail during product development owing to factors such as manufacturability, stability, lack of effectiveness, unacceptable levels of tolerability or safety concerns.

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Challegnes with adjuvant

- Lower reactogenicity adjuvants came with reduced vaccine efficacy.

- Safety and tolerability are critical regulatory issues facing the adjuvants.
- In most cases increased adjuvant potency is associated with increased reactogenicity and toxicity.

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Vaccine production

Steps of vaccine production

1. Generation of the Antigen VIRUS BACTERIA
2. Isolation of the Antigen
3. Purification
4. Addition of Other Components
5. Packaging

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Generation of the Antigen VIRUS BACTERIA

- The first step in order to produce a vaccine is generating the antigen that will trigger the immune response. For this purpose the pathogen's proteins or DNA need to be grown and harvested using the following mechanisms:
- Viruses are grown on primary cells such as cells from chicken embryos or using fertilised eggs (e.g. influenza vaccine) or cell lines that reproduce repeatedly (e.g. hepatitis A)
- Bacteria are grown in bioreactors which are devices that use a particular growth medium that optimises the production of the antigens
- Recombinant proteins derived from the pathogen can be generated either in yeast, bacteria or cell cultures.

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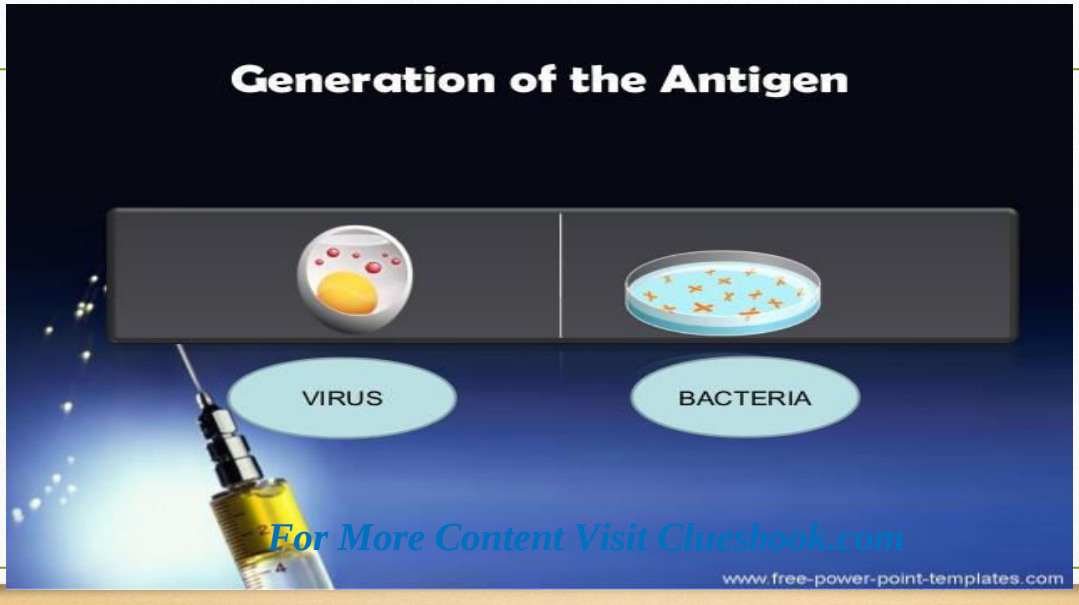

Isolation of the Antigen

- The aim of this second step is to release as much virus or bacteria as possible.
- To achieve this, the antigen will be separated from the cells and isolated from the proteins and other parts of the growth medium that are still present.

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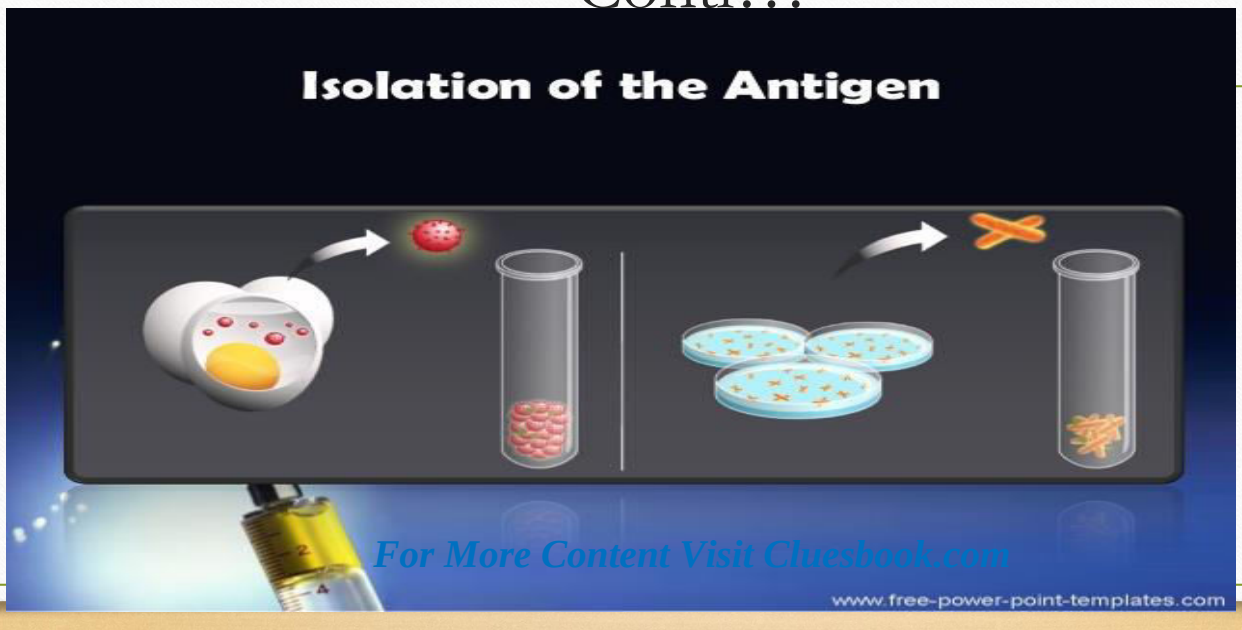

Purification

- In a third step the antigen will need to be purified in order to produce a high purity/quality product.
- This will be accomplished using different techniques for protein purification. For this purpose several separation steps will be carried out using the differences in for instance protein size, physico-chemical properties, binding affinity or biological activity.

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Addition of Other Components

- The fourth step may include the addition of an adjuvant, which is a material that enhances the recipient's immune response to a supplied antigen. The vaccine is then formulated by adding stabilizers to prolong the storage life or preservatives to allow multi-dose vials to be used safely as needed.
- Due to potential incompatibilities and interactions between antigens and other ingredients, combination vaccines will be more challenging to develop. Finally, all components that constitute the final vaccine are combined and mixed uniformly in a single vial or syringe.

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Packaging

- Once the vaccine is put in recipient vessel (either a vial or a syringe), it is sealed with sterile stoppers.
- All the processes described above will have to comply with the standards defined for Good Manufacturing Practices that will involve several quality controls and an adequate infrastructure and separation of activities to avoid cross-contamination.

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Microbial mining

- The use of microorganisms to remove minerals particularly metals from rocks
- **Biomining** is a technique of extracting metals from ores and other solid materials typically using prokaryotes or fungi.
- Recent progress in the genetic manipulation of microorganisms for industrial purposes promises to revitalize not only the bacterial leaching of metal-bearing ores but also the microbiological treatment of metal-contaminated waste water.

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- **The harsh, toxic** conditions in which metals are extracted from the earth may not produce a hospitable environment for humans, but for some microorganisms it is a suitable condition.
- Reveling in slurries that are often extremely acidic or basic, these microbes swim around, deriving energy from oxidation and reduction reactions that release economically valuable metals from minerals.

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- bacteria and archaea can be used to extract metal from rock as well as from retired mine sites whose toxic waste pools may still contain trace amounts of valuable metals.

- Harnessing microbes to do mining work is called biomining, or sometimes bioextraction or bioleaching.

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- Environmental contamination by heavy metals from anthropogenic and industrial activities has caused considerable irreparable damage to aquatic ecosystems.

- Several different physicochemical and biological processes are commonly employed to remove heavy metals from industrial wastewaters before their discharge into the environment

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- The bacteria involved in the leaching of metals from ores are among the most remarkable life forms known.

- The microorganisms are said to be chemolithotrophic ("rock-eating"); they obtain energy from the oxidation of inorganic substances.

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Bioleaching

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- the microorganisms in the river water were breaking down the sulfide-containing ores to extract copper

- microorganisms there played a role in the dissolution of metals from their ores, a process termed “bioleaching”.

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Bio-oxidation

- Bio-oxidation is another method in which microorganisms help to extract a metal of interest.
- However, during bio-oxidation the metal of interest remains in an insoluble form

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- Hence, bio-oxidation is often used as a pretreatment method prior to metal extraction, such as the extraction of ores containing gold.
- After bio-oxidation, the gold is extracted by a more traditional method such as cyanide extraction.
- cyanide extraction is the most commonly used leaching process for gold extraction

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biofuels

- A **biofuel** is defined as any fuel whose energy is obtained through a process of biological carbon fixation.
- Biofuels are fuels produced directly or indirectly from organic material biomass , including plant materials and animal waste.
- **Bioenergy** is energy derived from biofuels.
- Overall, bioenergy covers approximately 10% of the total world energy demand.

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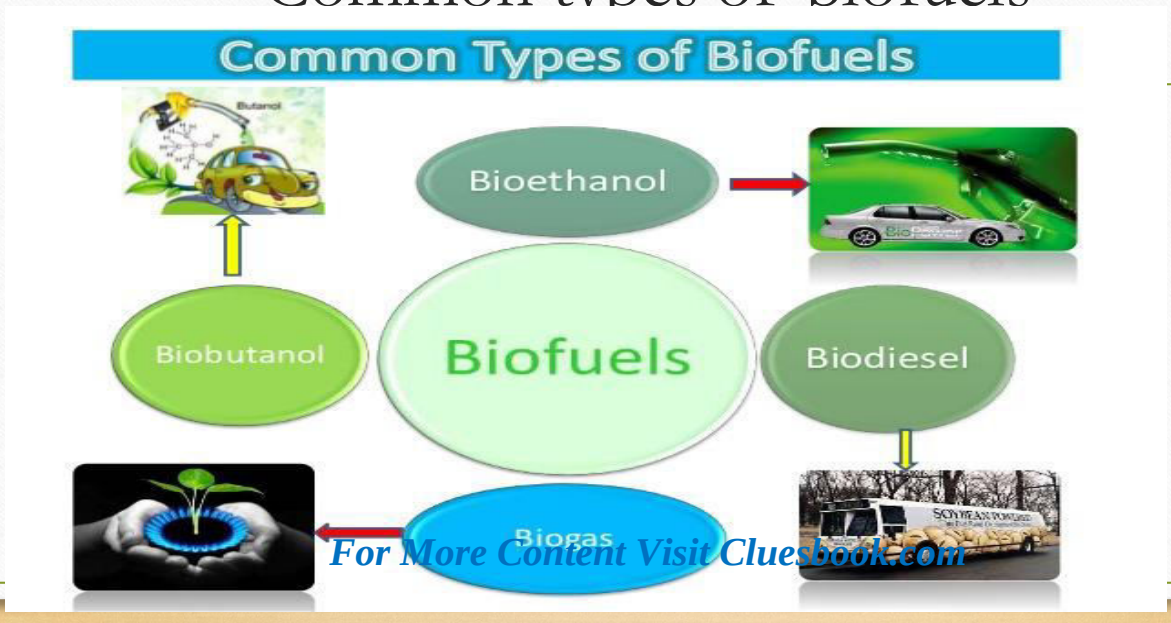
Common types of biofuels



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bioethanol

- Bio Ethanol is a type of alcohol that can be produced using any feedstock containing significant amounts of sugar, such as sugar cane or sugar beet, or starch such as maize and wheat.
- Sugar can be directly fermented to alcohol, while starch first needs to be converted to sugar.
- The fermentation process is similar to that used to make wine or beer, and pure ethanol is obtained by distillation

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Biodiesel

- **Biodiesel** is produced, mainly in the European Union, by combining vegetable oil or animal fat with an alcohol
- Biodiesel can be blended with traditional diesel fuel or burned in its pure form in compression ignition engines. Its energy content is somewhat less than that of diesel (88 to 95%).
- Biodiesel can be derived from a wide range of oils, including rapeseed, soybean, palm, coconut or jatropha oils and therefore the resulting fuels can display a greater variety of physical properties thanethanol

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biobutanol

- The term biobutanol refers to butanol made from renewable resources such as grain or cornstalks by fermentation process
- Bacteria; known as, solventogenic Clostridia is used
- Butanol is more similar to gasoline than to ethanol.

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biogas

- A mixture of CH4 , CO2 and other gases such as
- Carbon dioxide, Nitrogen ,Hydrogen ,Hydrogen sulfide ,Oxygen
- Biogas is a fuel used as domestic purpose
- Obtained from cow manure, fruit and vegetable waste
- Biogas is produced by the breakdown of organic waste by bacteria without oxygen anaerobic digestion

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First-generation

- **First-generation**biofuels are made from sugar, starch, vegetable oil, or animal fats using conventional technology. These are generally produced from grains high in sugar or starch fermented into bioethanol
- or seeds that which are pressed into vegetable oil used in biodiesel. Common first-generation biofuels include vegetable oils, biodiesel, bioalcohols, biogas, solid biofuels, syngas.

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Second-generation

- **Second-generation** biofuels are produced from non-food crops, such as cellulosic biofuels and waste biomass (stalks of wheat and corn, and wood).
- Common second-generation biofuels include vegetable oils, biodiesel, bioalcohols, biogas, solid biofuels, and syngas.
- Research continues on second-generation biofuels including biohydrogen, biomethanol, DMF, Bio-DME, Fischer-Tropsch diesel, biohydrogen diesel, mixed alcohols and wood diesel.

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Third-generation

- **Third-generation** biofuels are produced from extracting oil of algae – sometimes referred to as "oilgae".
- Its production is supposed to be low cost and high-yielding – giving up to nearly 30 times the energy per unit area as can be realized from current, conventional 'first-generation' biofuel feedstocks

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1. Transportation

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- More than 30% of energy consumed in the United States is used for vehicle transportation. Across the globe, transport takes account of 24% of energy and more than 60% of absorbed oil. This means that over a third of oil is used to operate vehicles.

- In short, biofuel can be turned into a hydrogen steam that is meant to be used in adjoining fuel-cell. More major car brands have already invested in stations for biofuel-powered vehicles

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Energy Generation

- In addition to producing fuel for transportation, fuel cells have a power-generating application that is available for electricity.
- Biofuel can be used to generate power in backup systems where emission matters most. This includes facilities such as schools, hospitals and other forms located in residential areas

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Provide Heat

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- Bioheat has grown over the past few years. As the primary use of natural gas that comes from fossil fuel, the heat that comes from hydraulic fracturing will lead to the production of natural gas. While natural gas does not need to come from fossil material, it is also able to originate from the recently grown material.
- A majority of biofuel that is used for heating is substantial. As wood is the most practical method to heat, houses that use wood burning stoves rather than gas or electricity. A blend of biodiesel will reduce the emission of both nitrogen and sulphur dioxide.

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Charging Electronics

- According to scientists from Saint Luis University, a fuel cell was developed with cooking oil and sugar to generate electricity
- consumers will be able to use these cells instead of generating electricity.
- Consumers may be able to use fuel cells in place of batteries to charge anything from computers to cell phones

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Cooking

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- While kerosene is the most common ingredient to use for stoves and non-wick lanterns, biodiesel works just as great

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Create energy when fossil fuel runs out

- As the oil supply is starting to run out.
- This has caused us to question how fuel can be extracted without destroying the environment.
- Biofuel –will help the government create a stable method of producing energy that is cost-effective.

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Reduce cost and need for imported oil

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- More than 84% of the world's petroleum is used in the United States. Despite the increase in fuel demands, the U.S. has recently started to decrease the need since 2006. This allows biofuels to become the best factor in energy reduction.
- Analysts say that replacing imported oil with biofuel will help to stabilise the economy when oil is disrupted.

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role of microbes in petroleum industry

- Petroleum-based products are the major source of energy for industry and daily life. Leaks and accidental spills occur regularly during the exploration, production, refining, transport, and storage of petroleum and petroleum products.
- Release of hydrocarbons into the environment whether accidentally or due to human activities is a main cause of water and soil pollution
- Soil contamination with hydrocarbons causes extensive damage of local system since accumulation of pollutants in animals and plant tissue may cause death or mutations

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- Microbial processes play an important role in industry, the protection of the natural environment
- Several practical applications of microbial processes are listed as follows

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Bioremediation

- **Bioremediation** is a natural process that involves the use of biological entities to neutralize the contaminated site
- **Microbial** biodegradation is the use of **bioremediation** and biotransformation methods to harness the naturally occurring ability of **microbial** xenobiotic metabolism to degrade, transform or accumulate environmental pollutants, including hydrocarbons (e.g. oil), polychlorinated biphenyls (PCBs), polyaromatic hydrocarbons

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Biosurfactants

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- These are microbial-synthesized surface-active substances that allow for more efficient microbial biodegradation of hydrocarbons in bioremediation processes.
- There are two ways by which biosurfactants are involved in bioremediation

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- 1) Increase the surface area of hydrophobic water-insoluble substrates. Growth of microbes on hydrocarbons can be limited by available surface area of the water-oil interface. Emulsifiers produced by microbes can break up oil into smaller droplets, effectively increasing the available surface area.
- 2) Increase the bioavailability of hydrophobic water-insoluble substrates. Biosurfactants can enhance the availability of bound substrates by desorbing them from surfaces (e.g. soil) or by increasing their apparent solubility. Some biosurfactants have low critical micelle concentrations (CMCs), a property which increases the apparent solubility of hydrocarbons by sequestering hydrophobic molecules into the centres of micelles.

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Microbial Enhanced Oil Recovery (MEOR)

- One of the major concerns facing the oil industry today is the recovery of the large percentage of oil remaining unrecovered in mature and in nearly depleted oil fields.
- Microbes can influence and improve the oil recovery process by:

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- generating gases that increase reservoir pressure and decrease oil viscosity
- generating acids that dissolve rock, thus improving absolute permeability
- reducing permeability in channels
- producing bio-surfactants that decrease interfacial tension
- reducing oil viscosity by degrading long-chain saturated hydrocarbons

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Plant-microbe interactions

- Plant-microbe interactions describe a broad range of scientific studies concerning how microbes interact with plants at the molecular biology and molecular genetics level.
- Plants and microbes can have a variety of interactions including
- Symbiotic
- associative
- pathogenic,
- all of which impact plant productivity, stress tolerance and disease resistance.

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SYMBIOTIC ASSOCIATIONS WITH CYANOBACTERIA

- Cyanbobacteria are broadly distributed in nature and form symbiotic relationships with many different organisms.
- cyanobacteria enhance their survival by establishing an association with a biological partner.
- Azolla: is an aquatic fern that contains bilobed leaves attached to a stem and is found floating in freshwater.
- The cyanobacteria are found in a cavity between the ventral and dorsal epidermal layers of the leaf

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Associative relation of microbes and plants

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- **Root Associations**
- The enzyme system for nitrogen fixation is found only in the prokaryotes, and in the case of symbiotic nitrogen fixation in plants,
- there is considerable specificity between the legume symbiont and bacteria for the stable association.
- The plant provides the carbon and energy source for the bacteria to grow, while the bacteria fix nitrogen with the production of amino acids for plant growth.

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Pathogenic interactions

- **Bacterial Pathogens**
- The bacteria-producing diseases in plants generally display a number of hydrolyticenzymes for penetration of the plant surface, and growth in the plant is generally intercellular between the parenchyma cells.
- As the disease progresses in the plant, bacterial distribution may become systemic as a result of migration into the vascular tissue.
- In some cases, the bacteria release pectinase that hydrolyzes the plant cell walls and the plant collapses.

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bio-fertilizers

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- INTRODUTION A biofertilizer is a substance which contains living microorganisms, when applied to seed, plant surfaces, or soil, colonizes the rhizosphere or the interior of the plant and promotes growth by increasing the supply or availability of primary nutrients to the host plant.
- Bio-fertilizers add nutrients through the natural processes of nitrogen fixation, solubilizing phosphorus, and stimulating plant growth through the synthesis of growth-promoting substances

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Advantages of bio fertilizers

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- Bio fertilizers are usually eco-friendly and also guard the particular ecosystem in opposition to pollution

- Bio fertilizers eliminate people damaging parts from your earth which usually result in conditions inside the crops. Crops can be safeguarded in opposition to drought as well as other stringent ailments through the use of resource fertilizers.

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Contii..

- Bio fertilizers usually are not expensive and also weak farmers can easily utilize these.
- They help acquire large produce regarding vegetation simply by creating the particular earth abundant together with vitamins and minerals and also microbes required for the particular progress with the crops
- Bio fertilizer increases the actual and also substance attributes regarding earth.

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Types of bio fertilizers

- **Nitrogen bio fertilizers**
- The sort of bio fertilizers can help any farmers to determine the nitrogen point during the solid ground. Nitrogen may be a crucial factor which happens to be put to use in any growing for put.
- Including, Azotobacteria must be used to your non-legume bounty; Rhizobium it takes to your legume bounty. Purple earth-friendly algae are needed to progress hemp despite the fact that Acetobacter must be used to progress sugarcane

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Phosphorus bio fertilizers

- Phosphorus bio fertilizers widely used to determine the phosphorus point during the solid ground. The decision for phosphorus to your put growing is small. Phosphorus bio fertilizers come up with any solid ground obtain demanded sum of phosphorus.

- Isn't crucial that your particular selected phosphorus bio fertilizers must be used for that selected types of head. They are put to use in all different kinds of any head including; Acetobacter, Rhizobium can implement.

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Compost bio fertilizers

- Compost bio fertilizers really are those that use the pet animal dung to make sure you greatly enhance all the dirty by means of advantageous microorganisms and additionally vitamin supplements.

- To make sure you replace the dog waste matter right into bio fertilizers, all the microorganisms want unhealthy bacteria proceed through organic technique and additionally help in breaking down all the waste matter. Cellulytic fungal civilization and additionally Azetobacter civilization can be installed for those compost bio fertilizers.

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Application methods of bio fertilizers

- Seedling root dip ,
- This method is applied to the rice crop. A bed of water is spread on the land where the crop has to grow. The seedlings of rice are planted in the water and kept there for eight to ten hours.

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Seed treatment

- Within this process your nitrogen along with phosphorus fertilizers are generally put together jointly throughout normal water. And then vegetables are generally dropped within this mix. Following purposes on this substance on the vegetables, vegetables are generally dried up. As soon as they normally dry out, they must always be sown immediately ahead of that they find harmed by simply unsafe bacteria

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Soil treatment

- All the bio fertilizers along with the compost fertilizers are mixed together.
- They are kept for one night. Then the next day the mixture is spread on the soil where seeds have to be sown

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biopesticides

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- Biopesticides are certain types of pesticides derived from such natural materials as animals, plants, bacteria, and certain minerals.
- Biopesticide is a formulation made from naturally occurring substances that controls pests by non toxic mechanisms and in ecofriendly manner.

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Advatages of biopesticides

- Inherently less harmful and less environmental load,
- Designed to affect only one specific pest or, in some cases, a few target organisms,
- Often effective in very small quantities and often decompose quickly, thereby resulting in lower exposures and largely avoiding the pollution problems .
- When used as a component of Integrated Pest Management (IPM) programs, biopesticides can contribute greatly.

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Types of biopesticides

- **Microbial pesticides**
- Microbial pesticides are composed of microscopic living organisms (viruses, bacteria, fungi, protozoa, or nematodes) or toxin produced by these organisms

• Applied as conventional insecticidal sprays, dusts, or granules.

• Their greatest strength is their specificity as most are essentially nontoxic and non pathogenic to animals and humans.

• Microbial pesticides includes insecticides, fungicides, herbicides and growth regulators of microbial origin.

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Plant-incorporated-protectants (PIPs)

- Pesticidal substances that plant produce from the genetic material that has been added to the plant.
- As the pest feed on such plants they will eventually die. Botanical pesticides:

• These are naturally occurring plant material that may be crude preparation of the plant parts ground to produce a dust or powder that can be used in full strength or dilute form in a carrier such as clay, talc or diatomaceous earth.

• "Azadirachtin" effects the reproductive and digestive procees of pest.

• Several plant based insecticides as nicotinoids, natural pyrethrins, rotenoids, neem products etc are used.

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Biochemical pesticides

- They are naturally occurring substance to control pest by non-toxic mechanisms.

• Biochemical pesticides include substances as insect sex pheromones, that interfere with mating that attract insect pest to traps

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PPT Slides 1 TO 50

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Regards : Zarva Chaudhary

Admin:

* Zarva Chaudhary *

Chaudhary Moazzam

* Laiba Mahi *

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What is compost

- Compost is organic matter that has been decomposed and recycled as a fertilizer and soil amendment
- Compost is a key ingredient in organic farming.
- End product of the decomposition of organic matter

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composting

- Composting is nature's process of recycling decomposed organic materials into a rich soil known as compost
- Composting is a natural process that turns organic material into a dark rich substance

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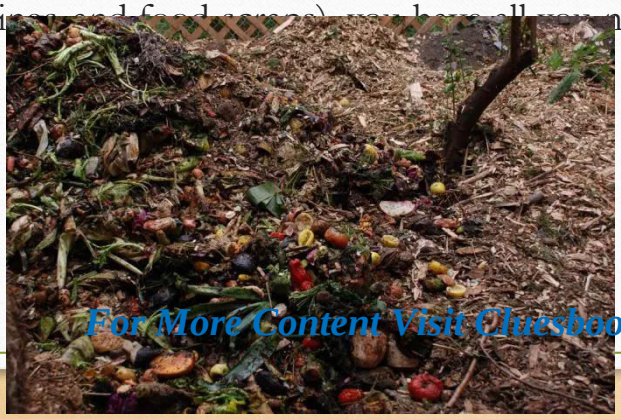
Types of composting

- **Backyard composting**
- f you have a yard and a balance of browns (fallen leaves or straw) and greens (grass clippi[illegible] need to make compost.



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Worm composting

Worm composting is using worms to recycle food scraps and other organic material into a valuable soil amendment called vermicompost, or worm compost.

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Grasscycling

- **Grasscycling** refers to an aerobic (requires air) method of handling

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Composting Benefits

- **Soil conditioner**: With compost, you are creating rich humus for lawn and garden. This adds nutrients to your plants and helps retain moisture in the soil.
- **Recycles kitchen and yard waste**: Composting can divert as much as 30% of household waste away from the garbage can.
- **Introduces beneficial organisms to the soil**: Microscopic organisms in compost help aerate the soil, break down organic material for plant use and ward off plant disease.

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- **Good for the environment**: Composting offers a natural alternative to chemical fertilizers.
-
- **Reduces landfill waste**: Most landfills in North America are quickly filling up; many have already closed down. One-third of landfill waste is made up of compostable materials

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antimicrobials

- Antimicrobial products kill or slow the spread of microorganisms. Microorganisms include bacteria, viruses, protozoans, and fungi such as mold and mildew.

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History

- The history of antimicrobials begins with the observations of Pasteur and Joubert, who discovered that one type of bacteria could prevent the growth of another.

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Uses of antimicrobials

- One recent study indicated that personal care products such as soaps, body washes and hand sanitizers that contain antimicrobial chemicals are more effective at limiting bacteria than soaps without antimicrobial agents – this 2011 study found that washing hands with an antimicrobial soap reduced bacteria on skin at a greater rate than non-antimicrobial soap

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Conti…

- Triclosan, an antimicrobial compound, has been found to promote oral health and aid in the fight against gingivitiswhen added to toothpaste.
- Kitchen counters, office desks, bathroom sinks, and other high-traffic areas in homes and offices may accumulate germs that can make people sick.
- Cleaning products without antibacterial/antimicrobial ingredients will clean the surface, but will not kill the germs on it

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Conti…

- Antimicrobial pesticides help to slow the growth of human pathogenic microorganisms and are used in food processing plants, dairies, breweries, poultry houses and other animal feeding operations, hospitals and medical and dental clinics and offices, municipal drinking water and water treatment facilities, swimming pools and spas, schools, day-care centers, public access facilities and homes

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types of antimicrobials

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- **Antibacterials**
- Antibacterials are used to treat bacterial infections
- The drug toxicity to humans and other animals from antibacterials is generally considered low
- Antibacterials are among the most commonly used drugs and among the drugs commonly misused by physicians, for example, in viral respiratory tract infectionsly low.
- Prolonged use of certain antibacterials can decrease the number of gut flora which may have a negative impact on health

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Antifungals

- Antifungals are used to kill or prevent further growth of fungi.
- In medicine, they are used as a treatment for infections such as athlete's foot, ringworm and thrush and work by exploiting differences between mammalian and fungal cells.
- They kill off the fungal organism without dangerous effects on the host.

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Antivirals

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- Antiviral drugs are a class of medication used specifically for treating viral infections.
- Like antibiotics, specific antivirals are used for specific viruses.
- They are relatively harmless to the host and therefore can be used to treat infections.

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Antiparasitics

- Antiparasitics are a class of medications indicated for the treatment of infection by parasites, such as nematodes, cestodes, trematodes, infectious protozoa, and amoebae.
- Like all antimicrobials against intracellular microbes, they must kill the infecting pest without serious damage to the host

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significance of microbial biotechnology in the economic development of Pakistan

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- Biotechnology is generally defined as application of living systems based technologies to develop commercial processes and products.
- Over the last few decades, several fundamental discoveries in life sciences have given rise to Modern Biotechnology which is now one of the fastest growing areas of science; hence this century has rightly been termed as 'Century of Biology'

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Biotechnology and Agriculture

- Agricultural biotechnology offers efficient and cost-effective means to produce a diverse array of novel, value-added products and tools.
- It has the potential to increase food production, reduce the dependency of agriculture on chemicals, and lower the cost of raw materials, all in an environmentally friendly manner.

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Conti…

- The initial phase of a revolution in agriculture has already occurred. Large areas of genetically modified (GM) crops of soybeans, maize, cotton, and canola have been successfully grown in the Western Hemisphere.
- The main objectives of creating transgenic plants are attempts to engineer metabolic pathways for the production of tailor-made plant polymers or low molecular weight compounds, increased resistance towards pathogens and pesticides, improved food quality, and the production of polypeptides for pharmaceutical or technical use. Plant-made vaccines or antibodies are especially attractive as plants are free of human diseases, reducing screening costs for viruses and bacterial toxins

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Economical Applications for Pakistan

- Pakistan is an agriculture base state
- Agriculture remains the dominant sector of the economy

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Bt Cotton

- Pakistan is the world's fourth largest producer of cotton after China, the USA and India
- Pakistan offers a rapidly expanding market for insecticides and pesticides. The total market has expanded from Rs. 7.20 billions ($120m) in 1990 to Rs11.00 billions ($184m) in 2000. By the introduction of the Bt cotton in Pakistan could result in a 45-55 per cent reduction in insecticide use on cotton (Which is 85per cent of Rs11 billion). This would mean a benefit of about of about Rs 4.2 to Rs 5.40 billions apart from the favorable impact on the environment and increase in cotton yield

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Bt Cotton

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Bt Rice

- In Pakistan average yield of conventional rice per acre is around 30 – 45 maund or 1,235.00 – 1,852.00 Kg. Bt Rice in Pakistan can increase per acre yield from 20 to 30 per cent.
- By the introduction of the Bt rice in Pakistan could result in a substantial reduction in insecticide use on rice and better control over disease like Stem Borers and Blight. This would mean enormous benefit to rice growers and whole economy, apart from the favorable impact on the environment and increase in rice yield.

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Bt Rice

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Biofuel

- A biofuel is a fuel that is produced through contemporary biological processes, such as agriculture and anaerobic digestion, rather than a fuel produced by geological processes such as those involved in the formation of fossil fuels, such as coal and petroleum, from prehistoric biological matter.

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Conti…

- Another growth area in this century will be the development of alternatives to non-renewable resources especially fossil fuels. Biotechnology will provide answers through modified enzymes and microorgamisms that can turn abundant biomass into feedstocks for the production of synthetics, plastics, polymers and bio-fuels like **Ethanol**
- Therefore, production of ethanol can spark new capital investment and economic development in rural communities across Pakistan

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Biodiesel.

- Biodiesel is a renewable, environment friendly substitute for petro-diesel fuel. It is produced from edible and non-edible oils and animal fats. Normally biodiesel blends can be used in diesel engines without any alteration.

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Conti..

- There are a number of benefits to producing and using bio-fuels aside from the transportation fuel supply issue. First, bio-fuels are produced domestically, and the feed-stocks for them are grown domestically.
- This helps reduce our country's trade deficit and creates jobs in our country, both of which are good for our economy.

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Conti…

- Our agricultural community especially stands to benefit, since bio-fuels are made from crops and agricultural residues, providing options for new valuable crops and new uses for existing crops and residues. Producing our fuel domestically also improves our energy security

- we become less dependent on the strategic, political, and economic whim of other countries.

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Genomics

- Genome sequencing of viruses began in the late 1970s. The basic technique involved, the random sequencing of fragments by the Sanger dideoxy termination method, was proposed and applied successfully by Fred Sanger and associates to the complete sequencing of bacteriophage DNAs, notably that of phage λ in 1980 for the principle of the random shotgun method).

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Sequencing history

J. Craig Venter and his associates at The Institute of Genomic Research successfully used the random shotgun approach for the entire 1.8-megabase (Mb) genome of *Haemophilus influenzae* without first directly cloning and ordering large inserts

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Prokaryotic Genome

There is a large variation in the size of bacterial genomes. Compared with the *H. influenzae* genome (1.8 Mb), some obligate parasites, such as *Chlamydia trachomatis* (1.0 Mb) and *Mycoplasma genitalum* (0.6 Mb), have smaller genomes. On the other hand, the genome of *E. coli* (4.6 Mb) is much larger, and even larger genomes are found in *Pseudomonas aeruginosa* (6.3 Mb), *Streptomyces coelicolor* (8.7 Mb), and *Bradyrhizobium japonicum* (9.1 Mb). (With rare exceptions, most archeal genomes have a rather small size, between 1.5 and 3 Mb).

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Homology

- Homology means descent from a common ancestor, and this is usually predicted by comparing sequences using computer programs such as BLAST. We distinguish two kinds of homologs. Homologous proteins in two different organisms, where they developed independently of each other, are called orthologs..

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Paralogs

- In contrast, homologous proteins that exist in the same species, presumably created originally by gene duplication and often differentiated later in terms of functions, are called paralogs

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Proteomics

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Proteomic analysis requires first the separation of thousands of proteins or their fragments and then their identification. It is impossible to identify this many proteins and their fragments by using the traditional approaches, such as reactivity with specific antibodies or functional properties.

Thus, the advances in mass spectrometry, now almost the universal approach in identification, were the major factor that made proteomics possible. Previously, mass spectrometry, which measures the mass-to-charge ratio of ions in a vacuum required the "hard" ionization methods that fragmented the sample molecule.

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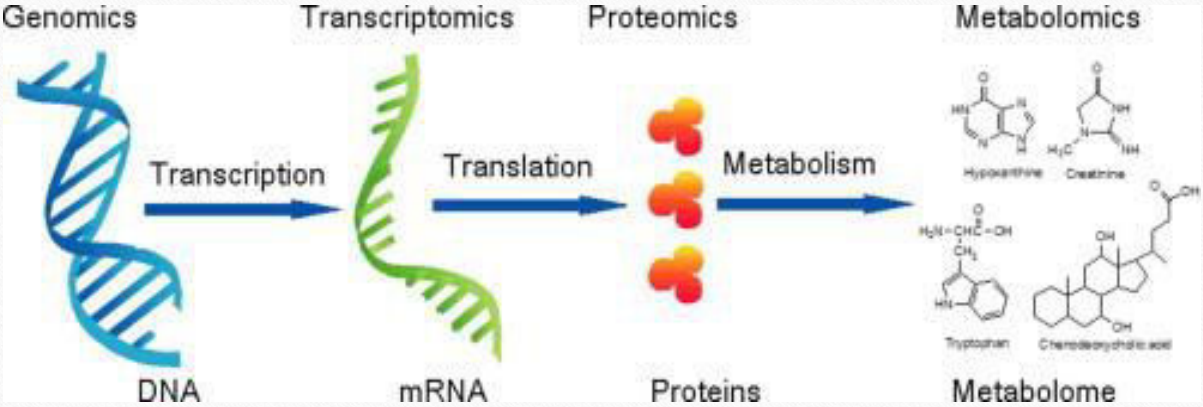
Flow chart



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Separation of proteins

- For the separation of proteins and peptides, two major approaches are used. In one, proteins are separated by two-dimensional polyacrylamide gel electrophoresis, usually by isoelectric points in the first dimension, and by molecular weight in the second dimension using sodium dodecyl sulfate SDS)-polyacrylamide gel electrophoresis

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- The protein spots are stained, and in this way they can be roughly quantitated. Each of the protein spots can then be digested with trypsin, and the fragments are analyzed by mass spectrometry. Commonly, MALDI is the ionization method used, and time-of-flight (TOF) detection is used. The molecular weights of various fragments are compared with their predicted sizes, obtained from the genomic database, and this results in the identification of the protein.
-

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Mass Spectrometry

- Mass spectrometry became extremely useful in the study of proteins and peptides when the two "soft ionization" methods – matrix-assisted laser desorption/ionization (MALDI) and electrospray ionization (ESI) were developed. These methods made possible the analysis of (unfragmented) peptides and proteins at high sensitivity and great precision. The significance of these advances may also be seen from the fact that two of the Nobel prizes in chemistry in 2002 were awarded to scientists who contributed to the development of these methods.

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MALDI

- MALDI is the abbreviation for "Matrix Assisted Laser Desorption/Ionization.“ The sample for MALDI is uniformly mixed in a large quantity of matrix. The matrix absorbs the ultraviolet light (nitrogen laser light, wavelength 337 nm) and converts it to heat energy. A small part of the matrix (down to 100 nm from the top outer surface of the Analyte in the diagram) heats rapidly (in several nano seconds) and is vaporized, together with the sample.

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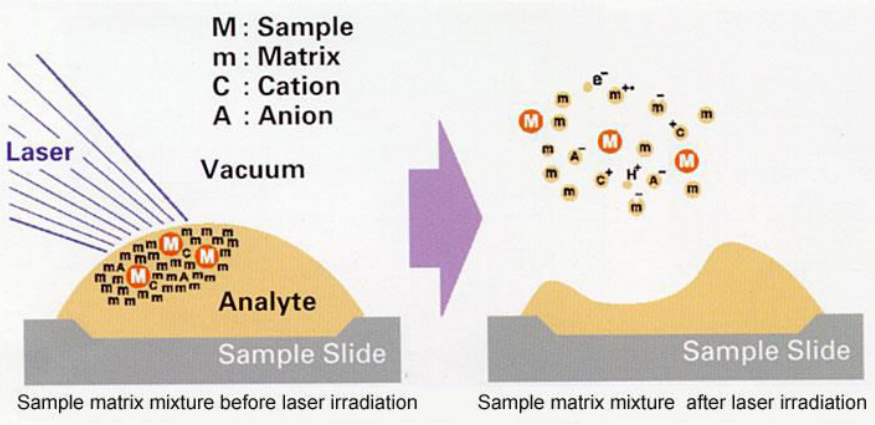
Diagram



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Metagenomics

- A fundamental operation in biotechnology is the expression of a foreign gene coding for a useful productive a suitable host organism, such as *E. coli* or yeast. However, because what is cloned is a piece of DNA, it may come directly from environmental sources. This approach of *direct cloning* is extremely valuable because most of the microorganisms in the environment have not yet been cultured, and the use of samples (for example soil samples) from a diverse range of environments is likely to supply genes that code for proteins with widely different properties.

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- Thus, it has now become a standard approach in the industry to maintain such samples for the direct cloning of useful genes. We can extend this approach, and instead of cloning just a gene or a gene cluster, we may hope to reconstruct the entire genomes of yet-to-be-cultured organisms from DNA samples in the environment. We will then be dealing not with the genome of a single cultured organism, but with genomes of the entire community of microorganisms, many of which may be currently uncultured. This is what is often called *metagenomics*.

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- Metabolomics, for example, supplies an exceptionally sensitive indicator of phenotypes, showing that metabolism becomes altered in mutant strains of yeast, in which conventional tests showed no alteration of phenotype. Metabolomics was also embraced eagerly by scientists working on plants, as plants produce a very large number of compounds

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Transcriptomics

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The presence of a gene in a genome does not obviously mean that it is expressed to produce mRNA and subsequently a protein. Furthermore, regulation of metabolism often involves alteration of expression of various genes, and therefore it is important to know their expression level in order to understand the physiology of the cell.

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Before the genome sequences became available, gene expression had to be examined on a one-by-one basis. The techniques used included Northern blot, which involves annealing of fractionated mRNA with the cognate pieces of DNA, and reverse transcriptase– PCR, in which mRNA is used as a template to produce cDNA, which is then amplified by PCR.

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Microarray

- When genome sequences became available, methods were developed in 1995 to "print" fragments of many genes onto a glass or plastic slide and to examine the binding of mRNA to these *microarrays*. The invention of this microarray or "chip" technology allowed us, for the first time, to examine the gene expression in an entire cell, for example, on an unbiased, "global" scale.

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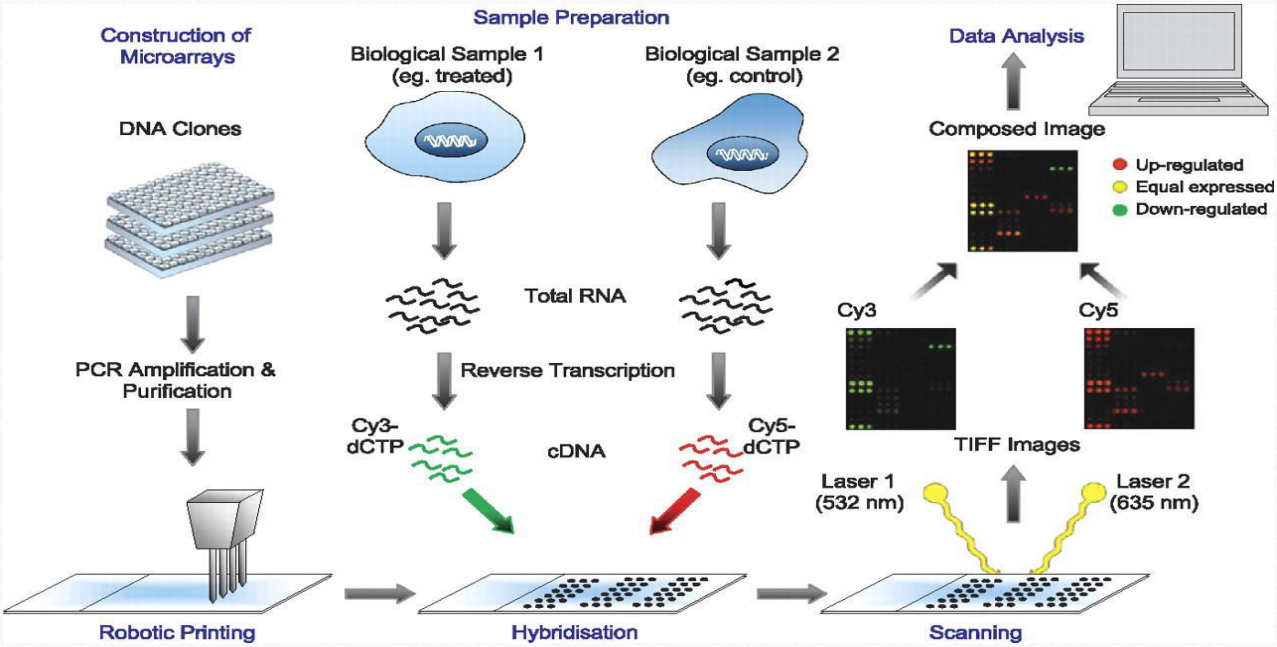

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Polyesters

- Bacteria differ from one another in the type of *reserve material* they accumulate when they are grown with unbalanced supplies of nutrients. The nature of the reserve material depends upon the genotype of the organism and the kind of limitation. If the carbon-to-nitrogen ratio is high, and if nitrogen, phosphorus, or oxygen are limiting, many bacteria will accumulate glycogen and/or aliphatic polyesters, polyhydroxyalkanoates (PHAs), in amounts up to 80% or more of their cellular dry weight.

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- In nature, *P. oleovorans* utilizes *n*-alkanes and *n*-alkanoic acids as sole carbon sources for growth. Under nutrient-limiting conditions, but with excess carbon source, *P. oleovorans* forms large amounts of poly(3-hydroxyalkanoates) from these substrates. *P. oleovorans* can use the "unnatural" substrate, 5- phenylvalerate (Figure 8.13), as the only carbon source. With this substrate in excess under conditions of nutrient deprivation, *P. oleovorans* forms a pure polyester, poly(3-hydroxy-5-phenylpentanoate) (Figure 8.13).

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Nutritional factors

- Nutritional requirements can be determined from chemical composition of microbial cells.

- Improper nutrients will lead to the formation of abnormal molecules and hinders the growth of microorganisms

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Issues with Microbial biotechnology

- **Physical factors**
- Temperature
- Humidity
- Ph
- viscosity

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Nutritional factors

- Macronutrients
- Micronutrients

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Nutritional factors

- Nutritional requirements can be determined from chemical composition of microbial cells.

- Improper nutrients will lead to the formation of abnormal molecules and hinders the growth of microorganisms

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Xantham gum

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- Xanthan gum was discovered in the mid-1950s, during a systematic search for useful biopolymers. In a screen of its large microbial culture collection, the Northern Utilization Research and Development Division of the U.S.Department of Agriculture discovered that the bacterium *Xanthomonas campestris* (originally isolated fromthe rutabaga plant) produced a polysaccharide with potentially valuable physical properties. Substantial commercial production of xanthan gum began in 1964, and in 1969 the U.S. FDA authorized its use in food. Today, xanthan gum has numerous uses, in the food industry and elsewhere.The total U.S. consumption of xanthan in 2005 is estimated at more than 80 million pounds.

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- The genus *Xanthomonas* belongs to the *γ* subdivision of the proteobacteria and consists exclusively of plant pathogens. It is a relative of the genus *Pseudomonas*, a group of bacteria discussed elsewhere in this book in the context of the degradation of xenobiotics.

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- *X. campestris* pathovar (abbreviated "pv.") *campestris*, the producer of xanthan gum, causes black rot, one of the most serious diseases of plants in the genus *Brassica*, which includes vegetables such as cabbage, cauliflower, brussels sprouts, broccoli, rutabaga, and turnips. Seeds contaminated with bacteria spread the infection. As the seedling emerges and grows, the bacteria colonize the surface of the plant.

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- In the case of *X. campestris* pv. *campestris*, the epiphytic bacteria enter the internal tissues through the *hydathodes*, structures on the leaf margin that allow the plant to excrete water. They then migrate through the vascular system and progressively cause chlorosis (destruction of chlorophyll), vein blackening (deposition of melaninlike pigments), and, ultimately, rotting.

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Primary structure

- The primary structure of xanthan gumis In common with cellulose, the xanthan backbone consists of β-(1→4)-linked d glucose; however, the 3-position of alternate glucose monomer units carries a trisaccharide side chain containing one glucuronic acid and two mannose residues. The nonterminal d-mannose unit carries an acetyl group at the 6- position, and a pyruvate is attached to the terminal d-mannose residue by a ketal linkage to the 4-and6-positions.Occasionally, the terminald-mannose carries a 6-*O*-acetyl group rather than pyruvate. The degree of acetylation and the pyruvate ketal content vary with culture conditions and from one strain of the microorganism to another.

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Good Luck

Regards : Zarva Chaudhary

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